

Report

Question 1: Water Jug Problem – Report

The Water Jug Problem was successfully solved using the state-space search approach. By performing fill, empty, and transfer operations, exactly **2 gallons** of water were obtained in the 4-gallon jug. This problem demonstrates how Artificial Intelligence solves problems by exploring different states systematically.

Question 2: Mars Rover Intelligent Agent – Report

The Mars Rover is a **Goal-Based Intelligent Agent** that uses sensors to perceive its environment and performs actions such as navigation, obstacle avoidance, and sample collection. The rover successfully completes its mission by making intelligent decisions based on the information collected from its surroundings.

Question 3: 8-Queens Problem – Report

The 8-Queens problem was solved using the **Backtracking Search Algorithm**. The algorithm places queens one by one in safe positions and backtracks whenever a conflict occurs. The final solution places all eight queens on the chessboard without any two queens attacking each other.

Question 4: OLA Cab Booking Problem – Report

The OLA Cab Booking system uses a **Goal-Based Agent** to help users select a suitable cab based on their destination and preferences. The program calculates the fare according to the selected cab type and confirms the booking. This demonstrates how AI agents make decisions to achieve a user's goal efficiently.

Question 5: Uniform Cost Search (UCS) – Report

The Uniform Cost Search algorithm successfully found the least-cost path from the source node **S** to the goal node **G**. The optimal path obtained was **S** → **A** → **C** → **G** with a minimum cost of **4**. UCS guarantees the optimal solution by always expanding the node with the lowest cumulative path cost.